

■ 2.2.2 Specifying Transformation Rules

There are many kinds of transformation rules built into *Mathematica*. These include rules for doing arithmetic, rules for simplifying algebraic expressions, rules for differentiation, and so on.

Perhaps *Mathematica*'s most powerful feature is that you can add your own rules to supplement, or even override, the ones that are built in. You can “teach” *Mathematica* about new kinds of functions and objects by giving it the rules it needs to manipulate them.

As Section 1.3 discussed, there is a growing library of packages which contain the rules needed for many different kinds of *Mathematica* applications.

We have already discussed two simple kinds of transformation rules that you can set up in *Mathematica*. When you define the value of a “variable”, say $x = 1 - b$, you are specifying the transformation rule that whenever x appears, it is to be replaced by $1 - b$. When you define a “function”, say $f[x_] = x^2$, you are specifying the transformation rule that any expression of the form $f[anything]$ should be replaced by $anything^2$.

Mathematica allows you to specify very general kinds of transformation rules for expressions. You can essentially define any expression to have any value.

This sets the expression $f[6]$ to have value 14. It specifies the rule that whenever $f[6]$ appears, it is to be replaced by 14.

```
In[1]:= f[6] = 14
Out[1]= 14
```

$f[6]$ is replaced according to the rule you have just defined. No rule is known for $f[7]$, so it is left unchanged.

```
In[2]:= a f[6] + b f[7]
Out[2]= 14 a + b f[7]
```

You can also specify a rule for $f[a]$.

```
In[3]:= f[a] = 8 a
Out[3]= 8 a
```

If you ask about f , *Mathematica* shows you all the rules that have been defined for it.

```
In[4]:= ?f
f
f/: f[6] = 14
f/: f[a] = 8 a
```

Now $f[a]$ is replaced using the rule that has been defined. There is no rule known for $f[b]$.

```
In[4]:= f[a] + f[b]
Out[4]= 8 a + f[b]
```

In the sections that follow, we will discuss how to define and use transformation rules for various kinds of expressions.

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